# Template Practices Document

## Contents



## Evaluated criteria

**C1** Example criterion description.
**C2** Example criterion description.
**C3** Example criterion description.

# 1 Worked Solution Examples

**Problems:**



## 1.1 Example Uniform Investment Plan (C1 C2)

A variable $V$ follows a uniform model on $[8, 16]$. Use the graph and density model to answer the tasks.

**Questions/Tasks.**

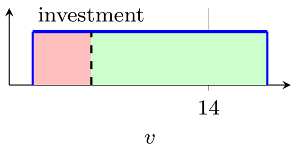
**C1** Write the density function and support.
**C2** Compute $P(10 \le V \le 14)$ and justify with area.

### C1 - Write the density function and support

$$f_V(v) = \begin{cases} \dfrac{1}{16-8} = \dfrac{1}{8}, & 8 \le v \le 16, \\ 0, & \text{otherwise.} \end{cases}$$

$$8 \le v \le 16 \qquad \longrightarrow \qquad f_V(v) = \frac{1}{8} \qquad \longrightarrow \qquad \int_8^{16} f_V(v)\, dv = 1$$

**C2 - Compute $P(10 \le V \le 14)$ and justify with area.**




$$P(10 \le V \le 14) = \int_{10}^{14} \frac{1}{8}\,dv = \frac{1}{8}(14-10) = \frac{4}{8} = 0.5$$

The shaded area represents half of the support interval.

## 1.2 Example Linear Density Model (C2)

Suppose $X$ has density $f(x) = kx$ for $0 \le x \le 2$.

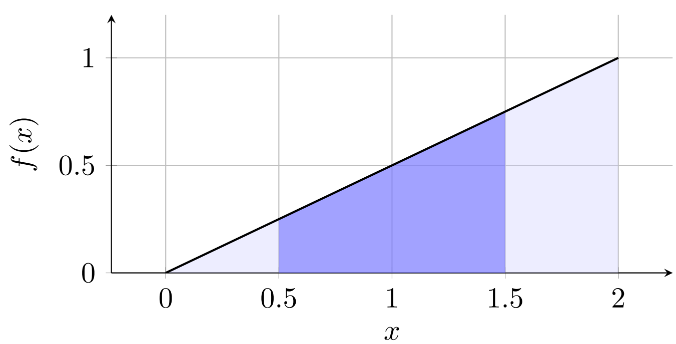
**Questions/Tasks.**

**C2.1** Find the value of $k$.

**C2.2** Compute $P(0.5 \le X \le 1.5)$.

**C2.1 - Find the value of $k$.**

$$1 = \int_0^2 kx\,dx = k\left[\frac{x^2}{2}\right]_0^2 = 2k \quad \Rightarrow \quad k = \frac{1}{2}$$




**C2.2 - Compute $P(0.5 \le X \le 1.5)$.**

$$P(0.5 \le X \le 1.5) = \int_{0.5}^{1.5} \frac{1}{2}x\,dx = \left[\frac{x^2}{4}\right]_{0.5}^{1.5} = \frac{1.5^2 - 0.5^2}{4} = 0.5$$

## 1.3 Example Table and Probability Summary (C3)

A summary table is shown for two intervals of a piecewise density.

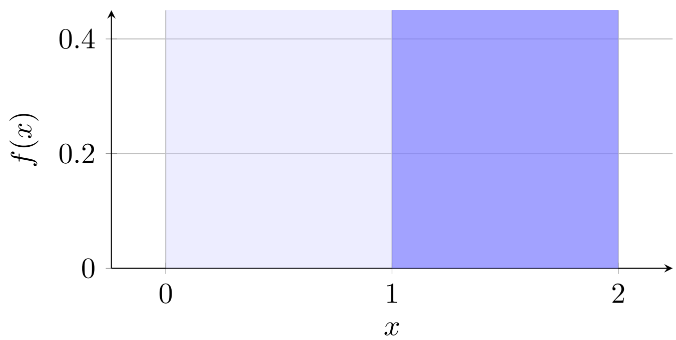
**Questions/Tasks.**

**C3.1** Complete the probability summary table.

**C3.2** State one consistency check.

**C3.1 - Complete the probability summary table.**

| Interval | Expression | Result |
|---|---|---|
| $[0, 1]$ | $\int_0^1 0.4\,dx$ | 0.4 |
| $[1, 2]$ | $\int_1^2 0.6\,dx$ | 0.6 |
| Total | $0.4 + 0.6$ | 1 |




**C3.2 - State one consistency check.**

A valid check is that all interval probabilities are nonnegative and add to 1.

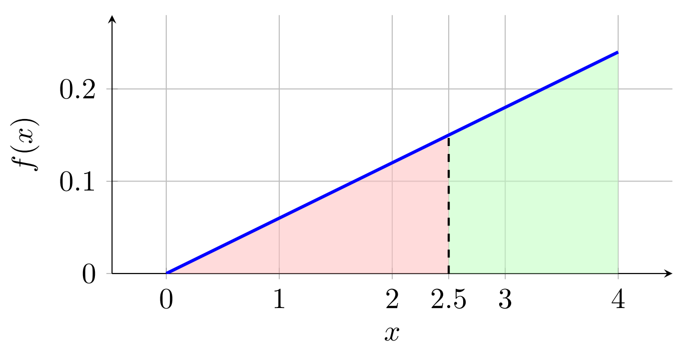
# 2 Graph Command Examples

This section provides one representative command from each graph file in `book/preamble/graphs`.

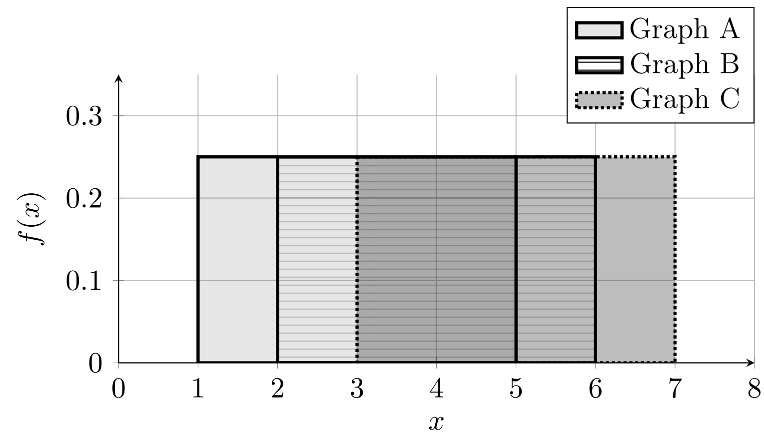
**Problems:**



## 2.1 Example from `uniform_linear_probability_distributions.tex`.



### 2.2 Example from `probability_distribution_graphs.tex`.




### 2.3 Example from `normal_distribution_graphs.tex`.